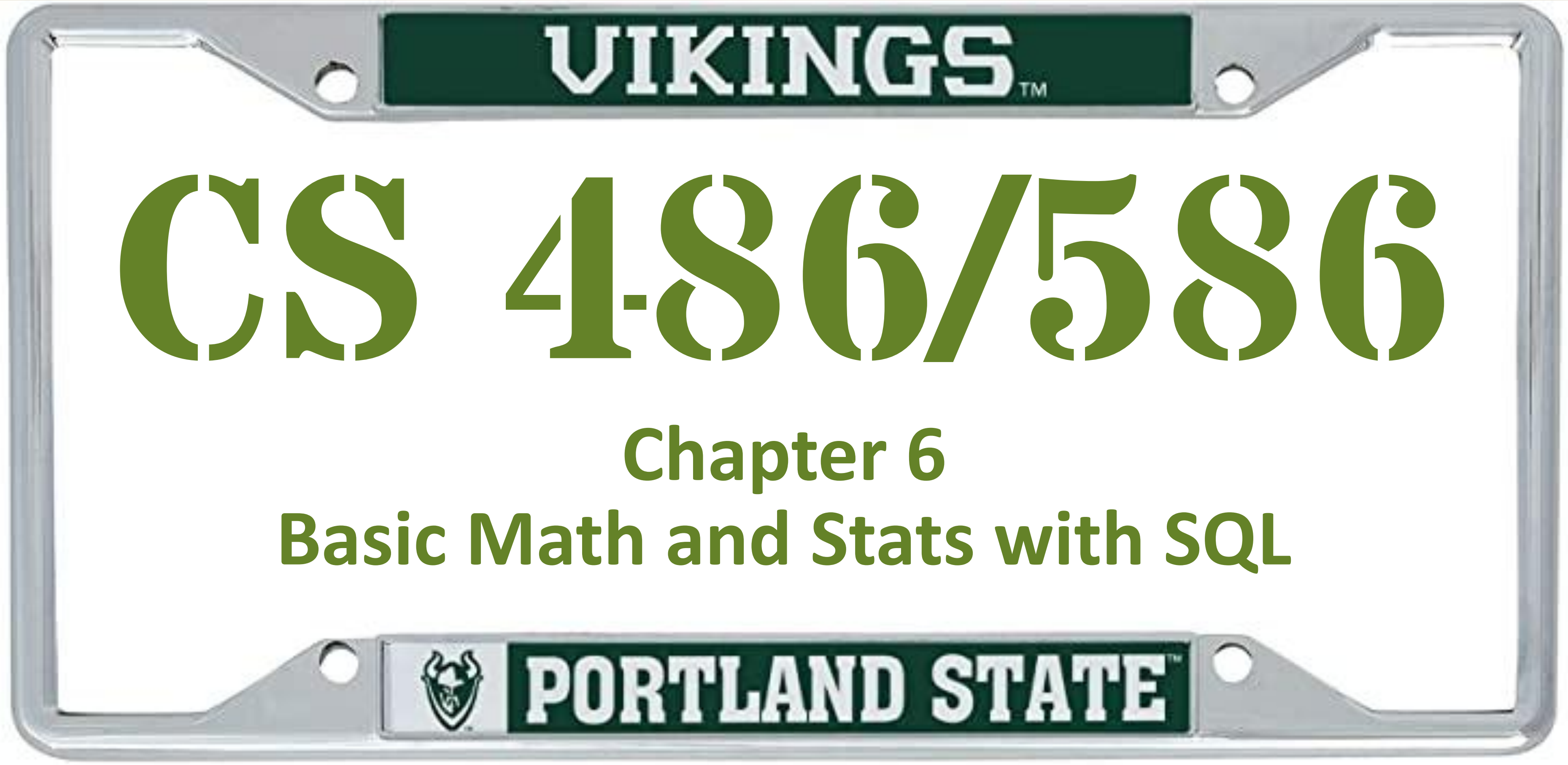

CS 486/586

Chapter 6
Basic Math and Stats with SQL

Operator	Description	Example	Result
+	addition	2 + 3	5
-	subtraction	2 - 3	-1
*	multiplication	2 * 3	6
/	division (**integer division truncates the result**)	4 / 2	2
%	modulo (**integral remainder**)	5 % 4	1
^	exponentiation (**Unlike typical mathematical practice, multiple uses of ^ will associate left to right by default**)	2.0 ^ 3.0	8
|/	square root (**also a SQRT() function**)	|/ 25.0	5
||/	cube root (**also CBRT() function**)	||/ 27.0	3
!	factorial (**deprecated, use factorial() instead**)	**5 !**	120
!!	factorial as a prefix operator (**deprecated, use factorial() instead**)	**!! 5**	120
@	absolute value (**also an ABS() function**)	@ -5.0	5
&	bitwise AND	91 & 15	11
|	bitwise OR	32 | 3	35
#	bitwise XOR	17 # 5	20
~	bitwise NOT (ones' complement)	~1	-2
<<	bitwise shift left	1 << 4	16
>>	bitwise shift right	8 >> 2	2

When using an operator between two numbers — addition, subtraction, multiplication, or division — the data type returned follows this pattern:

- Two integers return an integer.
- A numeric datatype on either side or both sides of the operator returns a numeric.
- Anything with a floating-point number returns a floating-point number of type `double precision`.

For the PostgreSQL operators discussed so far, the order of precedence is as follows:

1. Exponents and roots
2. Multiplication, division, modulo
3. Addition and subtraction

When in doubt

Parentheses are not endangered, use them.

(((((((((( ))))))))))

```sql
SELECT county_name AS county,
   state_name AS state, births_2019 AS births, deaths_2019 AS deaths,
   births_2019 - deaths_2019 AS natural_increase
FROM chap_05_examples.us_counties_pop_est_2019
WHERE state_name = 'Oregon'
ORDER BY state_name, county_name
LIMIT 10;
```

county	state	births	deaths	natural_increase
Baker County	Oregon	153	211	-58
Benton County	Oregon	701	584	117
Clackamas County	Oregon	3,987	3,559	428
Clatsop County	Oregon	377	398	-21
Columbia County	Oregon	499	503	-4
Coos County	Oregon	599	937	-338
Crook County	Oregon	259	266	-7
Curry County	Oregon	166	345	-179
Deschutes County	Oregon	1,857	1,548	309
Douglas County	Oregon	1,065	1,473	-408

```sql
SELECT county_name AS county,
    state_name AS state,
    area_water::numeric / (area_land + area_water) * 100 AS pct_water
FROM chap_05_examples.us_counties_pop_est_2019
where state_name = 'Oregon'
ORDER BY pct_water desc
limit 10;
```

county	state	pct_water
Clatsop County	Oregon	23.5748770871
Curry County	Oregon	18.1404696809
Lincoln County	Oregon	17.8770518328
Tillamook County	Oregon	17.2649189479
Coos County	Oregon	11.6414330358
Multnomah County	Oregon	7.3225547721
Columbia County	Oregon	4.4792851059
Lane County	Oregon	3.5651682396
Klamath County	Oregon	3.1687716021
Lake County	Oregon	2.6262498945

```sql
SELECT county_name AS county,
  state_name AS state,
  pop_est_2018 AS pop_2018,
  pop_est_2019 AS pop_2019,
  pop_est_2019 - pop_est_2018 as pop_growth,
  round((pop_est_2019 - pop_est_2018 * 1.0) / pop_est_2018 * 100.0, 4) as pct_pop_growth
FROM chap_05_examples.us_counties_pop_est_2019
where state_name = 'Oregon'
ORDER BY pct_pop_growth desc limit 10;
```

county	state	pop_2018	pop_2019	pop_growth	pct_pop_growth
Sherman County	Oregon	1,700	1,780	80	4.7059
Deschutes County	Oregon	191,905	197,692	5,787	3.0156
Crook County	Oregon	23,825	24,404	579	2.4302
Morrow County	Oregon	11,360	11,603	243	2.1391
Wallowa County	Oregon	7,058	7,208	150	2.1252
Jefferson County	Oregon	24,150	24,658	508	2.1035
Linn County	Oregon	127,451	129,749	2,298	1.803
Lincoln County	Oregon	49,224	49,962	738	1.4993
Harney County	Oregon	7,292	7,393	101	1.3851
Clatsop County	Oregon	39,711	40,224	513	1.2918

We've shown math operations across columns in **each row of a table.**
SQL also lets you calculate a result from values within the same column using aggregate functions.

Aggregate functions perform a calculation on a set of rows and return a single row. PostgreSQL provides all standard SQL's aggregate functions as follows:

- `AVG()` – return the average value.
- `COUNT()` – return the number of values (with or without `NULL`).
- `MAX()` – return the maximum value.
- `MIN()` – return the minimum value.
- `SUM()` – return the sum of all or distinct values.

We've shown math operations across columns in **each row of a table.**
SQL also lets you calculate a result from values within the same column using aggregate functions.

Below, we apply a couple **aggregate functions** to the **entire table**.

```
SELECT
    sum(pop_est_2019) AS county_sum,
    round(avg(pop_est_2019), 0) AS county_average
FROM chap_05_examples.us_counties_pop_est_2019;
```

county_sum	county_average
328,239,523	104,468